1 Баланс импульса частиц и поля

$$\frac{\partial w}{\partial t} + \operatorname{div}\bar{S} + (\bar{j}\bar{E}) = 0$$ — баланс эн. частиц и поля

$$W = \frac{\bar{E}^2 + \bar{H}^2}{8\pi}$$ пл-ть энергии ЭМ поля

$$\bar{S} = \frac{c}{4\pi}[\bar{E}, \bar{H}]$$ в-р Пойнтинга, пл-ть потока энергии ЭМ поля

$$\frac{\partial}{\partial t}\left(\frac{S_i}{c^2}\right) + \frac{\partial}{\partial x_j}\sigma_{ij} + \rho E_i + \frac{1}{c}[\bar{j}, \bar{H}]_i = 0$$

$$g_i = \frac{S_i}{c^2}, \quad \sigma_{ij} = w\delta_{ij} - \frac{1}{4\pi}(E_i E_j + H_i H_j)$$

инт. по всему пр-ву

$$\frac{\partial}{\partial t}\int dV g_i + \underbrace{\int dV \frac{\partial}{\partial x_j}\sigma_{ij}}_{\oint dS_j \sigma_{ij} = 0} + \underbrace{\int dV \rho E_i}_{\sum_a dV e_a \delta(\bar{r}-\bar{r}_a) E_i(\bar{r}) = \sum_a e_a E_a(\bar{r}_a)} + \underbrace{\int dV \frac{1}{c}[\bar{j}, \bar{H}]_i}_{\sum_a dV \frac{e_a}{c}[\bar{v}_a, \bar{H}(\bar{r})]_i \delta(\bar{r}-\bar{r}_a) = \sum_a \frac{e_a}{c}[\bar{v}_a, \bar{H}(\bar{r}_a)]_i} = 0$$

$$\frac{d\bar{P}_{кин}}{dt} = e\bar{E} + \frac{e}{c}[\bar{v}, \bar{H}]$$

$$\frac{d}{dt}\left(\int dV g_i + P_{кин\,i}\right) = 0$$

$$\to \bar{g} = \frac{\bar{S}}{c^2}$$ пл-ть импульса ЭМ поля

$$\int dV \bar{g}$$ полный импульс ЭМ поля

Инт. по пустому кон. объёму

$$\frac{\partial}{\partial t}\int_V dV g_i = -\oint dS_j \sigma_{ij}$$

σ_{ij} пл-ть потока i-компоненты импульса через площадь, ориент. по напр. j

$$\frac{\partial}{\partial t} g_i + \frac{\partial}{\partial x_j}\sigma_{ij} + \rho E_i + \frac{1}{c}[\bar{j}, \bar{H}]_i = 0$$ баланс имп. частиц и поля

Пустое пр-во

$$\frac{\partial w}{\partial t} + \operatorname{div}\bar{S} = 0$$

$$\frac{\partial}{\partial t} g_i + \frac{\partial}{\partial x_j}\sigma_{ij} = 0$$

$$\Rightarrow \frac{\partial T^{\mu\nu}}{\partial x^\nu} = 0, \quad T^{\mu\nu} = \begin{pmatrix} w & \frac{cS_i}{c} \\ \frac{S_i}{c} & \sigma_{ij} \end{pmatrix}$$ тензор энергии-импульса ЭМ поля

$$= \frac{1}{4\pi}\left(\frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta} - F^{\mu\alpha} F_\alpha{}^\nu\right)$$ (след = 0)

Баланс энергии-импульса частиц и поля

$$\frac{\partial T^{\mu\nu}}{\partial x^\nu} = -\frac{1}{c} F^{\mu\nu} j_\nu$$

2 Волновые ур-я для потенциалов

2 пара ур-й Максвелла:

$$\operatorname{div}\bar{E} = 4\pi\rho$$

$$\operatorname{rot}\bar{H} = \frac{1}{c}\frac{\partial \bar{E}}{\partial t} + \frac{4\pi}{c}\bar{j}$$

$$\bar{E} = -\frac{1}{c}\frac{\partial \bar{A}}{\partial t} - \frac{\partial \varphi}{\partial \bar{r}} \qquad \Rightarrow \qquad -\Delta\varphi - \frac{1}{c}\frac{\partial}{\partial t}\operatorname{div}\bar{A} = 4\pi\rho$$

$$\bar{H} = \operatorname{rot}\bar{A} \qquad \operatorname{grad}\operatorname{div}\bar{A} - \Delta\bar{A} = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\bar{A} - \frac{1}{c}\frac{\partial}{\partial t}\frac{\partial \varphi}{\partial \bar{r}} + \frac{4\pi}{c}\bar{j}$$

Лор. калибр.: $\frac{1}{c}\frac{\partial \varphi}{\partial t} + \operatorname{div}\bar{A} = 0$ $\left(\frac{\partial A^\nu}{\partial x^\nu} = 0\right)$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\varphi = 4\pi\rho$$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\bar{A} = \frac{4\pi}{c}\bar{j}$$

4Д

$$\frac{\partial F^{\mu\nu}}{\partial x^\nu} = -\frac{4\pi}{c} j^\mu$$

$$\frac{\partial}{\partial x^\nu}\left(\frac{\partial A^\nu}{\partial x_\mu} - \frac{\partial A^\mu}{\partial x_\nu}\right) = -\frac{4\pi}{c} j^\mu$$

$$\frac{\partial}{\partial x_\mu}\left(\frac{\partial A^\nu}{\partial x^\nu}\right) - \frac{\partial^2 A^\mu}{\partial x^\nu \partial x_\nu} = -\frac{4\pi}{c} j^\mu$$

$\frac{\partial A^\nu}{\partial x^\nu} = 0$

$$\frac{\partial^2}{\partial x^\nu \partial x_\nu} A^\mu = +\frac{4\pi}{c} j^\mu$$

$$\Box A^\mu = +\frac{4\pi}{c} j^\mu$$

Даламбертиан

Кулоновская калибр.: $\operatorname{div}\bar{A} = 0$

$$\Delta\varphi = -4\pi\rho$$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\bar{A} = -\frac{1}{c}\frac{\partial}{\partial t}\frac{\partial \varphi}{\partial \bar{r}} + \frac{4\pi}{c}\bar{j}$$

3 Электростатика

$\bar{V}_a = 0$, $\bar{j}_a = 0$, $\frac{\partial}{\partial t} = 0$, $\bar{H} = 0$

$\operatorname{div}\bar{E} = 4\pi\rho$, $\bar{E} = -\frac{\partial \varphi}{\partial \bar{r}}$

$$\Delta\varphi = -4\pi\rho$$

+ 1 точечный заряд в начале коорд.



$\rho = e\delta(\bar{r})$

$\oint d\bar{S}\bar{E} = 4\pi e$

$\oint d\bar{S}\bar{E} = 4\pi e$ $\qquad \bar{E} \parallel \bar{r}$

$E \cdot 4\pi r^2 = 4\pi e$ $\Rightarrow$ $E = \frac{e}{r^2}$, $\bar{E} = \frac{e\bar{n}}{r^2} = \frac{e\bar{r}}{r^3}$

$\varphi = \frac{e}{r} + \underset{\parallel}{\underset{0}{\text{const}}}$ $\qquad (\varphi(\infty) = 0)$

$$\varphi = \frac{e}{|\bar{r} - \bar{r}_a|}$$



$$\bar{E} = \frac{e(\bar{r} - \bar{r}_a)}{|\bar{r} - \bar{r}_a|^3}$$

$e, \bar{r}_a$

$$\varphi(\bar{r}) = \sum_a \frac{e_a}{|\bar{r} - \bar{r}_a|}, \quad \bar{E}(\bar{r}) = \sum_a \frac{e_a(\bar{r} - \bar{r}_a)}{|\bar{r} - \bar{r}_a|^3}$$

$$\varphi(\bar{r}) = \int d^3\bar{r}' \frac{\rho(\bar{r}')}{|\bar{r} - \bar{r}'|} \leftarrow \sum_a e_a \delta(\bar{r} - \bar{r}_a)$$

$$\bar{E}(\bar{r}) = \int d^3\bar{r}' \frac{\rho(\bar{r}')(\bar{r} - r')}{|\bar{r} - \bar{r}'|^3}$$

4 Ф-я Грина ур-я Пуассона

$$\Delta\varphi = -4\pi\rho$$

$$\Delta_{\bar{r}} G(\bar{r} - \bar{r}') = \delta(\bar{r} - \bar{r}')$$

ф-я Грина

рем. ур-я Пуассона $\varphi(\bar{r}) = \int d^3\bar{r}' (-4\pi\rho(\bar{r}')) G(\bar{r} - \bar{r}')$ (+const)

$$\Delta_{\bar{r}} \varphi(\bar{r}) = \Delta_r \int d^3\bar{r}' (-4\pi\rho(\bar{r}')) G(\bar{r} - \bar{r}') = \int d^3\bar{r}' (-4\pi\rho(\bar{r}') \Delta_{\bar{r}} G(\bar{r} - \bar{r}')) = \int d^3\bar{r}' (-4\pi\rho(\bar{r}'))$$

$$\cdot \delta(\bar{r} - \bar{r}') = -4\pi\rho(\bar{r})$$

$$\bar{R} = \bar{r} - \bar{r}', \quad \Delta_{\bar{r}} = \Delta_{\bar{R}}$$

$$\Delta_{\bar{R}} G(\bar{R}) = \delta(\bar{R})$$

$$G(\bar{R}) = \frac{1}{(2\pi)^3} \int d^3\bar{k}\, e^{i\bar{k}\bar{R}} \tilde{G}(\bar{k})$$

$$\delta(\bar{R}) = \frac{1}{(2\pi)^3} \int d^3\bar{k} e^{i\bar{k}\bar{R}} \cdot 1$$

$$\Delta_{\bar{R}} G(\bar{R}) = \frac{1}{(2\pi)^3} \int d^3\bar{k}\, \Delta_{\bar{R}} (e^{i\bar{k}\bar{R}}) \tilde{G}(\bar{k}) =$$

$$\Delta(e^{i\bar{k}\bar{R}}) = \partial_i \partial_i e^{ik_j R_j} = \partial_i (e^{ik_j R_j} \partial_i (ik_j R_j)) = ik_i \partial_i (e^{ik_j R_j}) = -\bar{k}^2 e^{i\bar{k}\bar{R}}$$

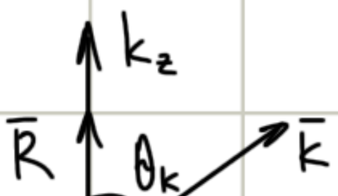
$$= \frac{-1}{(2\pi)^3} \int d^3\bar{k}\, k^2 \tilde{G}(\bar{k}) e^{i\bar{k}\bar{R}}$$

$$\frac{1}{(2\pi)^3} \int d^3k\, e^{i\bar{k}\bar{R}} (-k^2 \tilde{G}(\bar{k}) - 1) = 0$$

$$\forall \bar{R} \Rightarrow \tilde{G}(\bar{k}) = -\frac{1}{k^2}$$

$$G(\bar{R}) = \frac{-1}{(2\pi)^3} \int d^3k \frac{e^{i\bar{k}\bar{R}}}{k^2}$$

$$G(\bar{R}) = \frac{-1}{(2\pi)^3} \int dk\, k^2 \int_0^{\pi} d\theta_k \sin\theta_k \int_0^{2\pi} d\varphi_k \frac{e^{ikR\cos\theta_k}}{k^2} = -\frac{1}{(2\pi)^2 R} \int_0^{\infty} d(kR) \left[\frac{e^{ikR} - e^{ikR(-1)}}{ikR} \right] = -\frac{1}{2\pi^2 R} \int_0^{\infty} \frac{\sin x}{x} dx =$$



$$= -\frac{1}{4\pi R} = -\frac{1}{4\pi|\bar{R}|}$$

$$\bar{G}(\bar{r} - \bar{r}') = \frac{-1}{4\pi|\bar{r} - \bar{r}'|}$$

ф-я Грина ур-я Пуассона (ЭП)

$$\Delta\varphi = -4\pi\rho$$

$$\varphi(\bar{r}) = \int d^3\bar{r}' (-4\pi\rho(\bar{r}')) G(\bar{r} - \bar{r}') = \int d^3\bar{r}' \frac{\rho(\bar{r}')}{|\bar{r} - \bar{r}'|}$$

$$\bar{E}(\bar{r}) = \int d^3\bar{r}' \frac{\rho(\bar{r}')(\bar{r} - \bar{r}')}{|\bar{r} - \bar{r}'|^3}$$

$$\Delta\varphi = -4\pi\rho$$

1 мое. изм. $\rho = e\delta(\bar{r} - \bar{r}_0)$

$$\varphi(\bar{r}) = \frac{e}{|\bar{r} - \bar{r}_0|}$$

$$\Delta\left(\frac{1}{|\bar{r} - \bar{r}_0|}\right) = -4\pi\delta(\bar{r} - \bar{r}_0)$$

$$\Delta\left(\frac{1}{r}\right) = -4\pi\delta(\bar{r})$$